

Task 4: Session 4 (AIML) - Assignment Questions

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Domain: AIML

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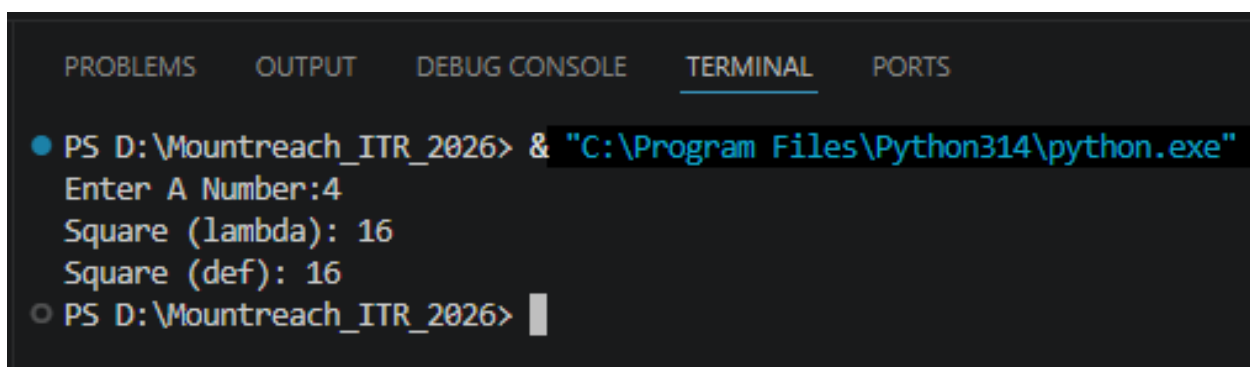
1. Lambda Function

Write a lambda function named square that takes a number x and returns its square.

Call the lambda function with input from the user and print the result.

Also write the same logic using a normal def function and compare both in comments.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A Number:4
Square (lambda): 16
Square (def): 16
PS D:\Mountreach_ITR_2026>
```

Program Code:

```
# Lambda Function
# Lambda: square = lambda x: x ** 2 -> Single-Line, No Name Needed Internally
# Use lambda For Short, Throwaway Operations (e.g., Inside Map/Filter/Sorted).
square = lambda x: x ** 2

# Normal def Function For Comparison
# Def: def square_def(x): -> Multi-Line, Readable
# Use def For Anything Reusable, Complex, Or That Needs Documentation.
def square_def(x):
    return x ** 2

number = int(input("Enter A Number:"))

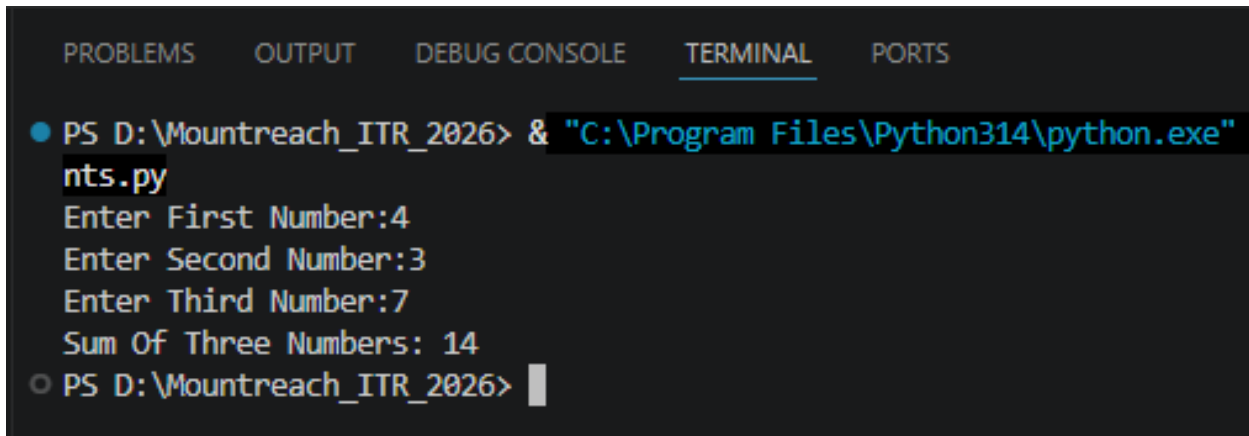
print("Square (lambda):", square(number))
print("Square (def):", square_def(number))
```

2. Lambda With Multiple Arguments

Create a lambda function `add_three(a, b, c)` that returns the sum of three numbers.

Take three numbers as input from the user and print their sum using this lambda function.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe" nts.py
Enter First Number:4
Enter Second Number:3
Enter Third Number:7
Sum Of Three Numbers: 14
○ PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
# Lambda Function With Multiple Arguments
add_three = lambda a, b, c: a + b + c

a = int(input("Enter First Number:"))
b = int(input("Enter Second Number:"))
c = int(input("Enter Third Number:"))
print("Sum Of Three Numbers:", add_three(a, b, c))
```

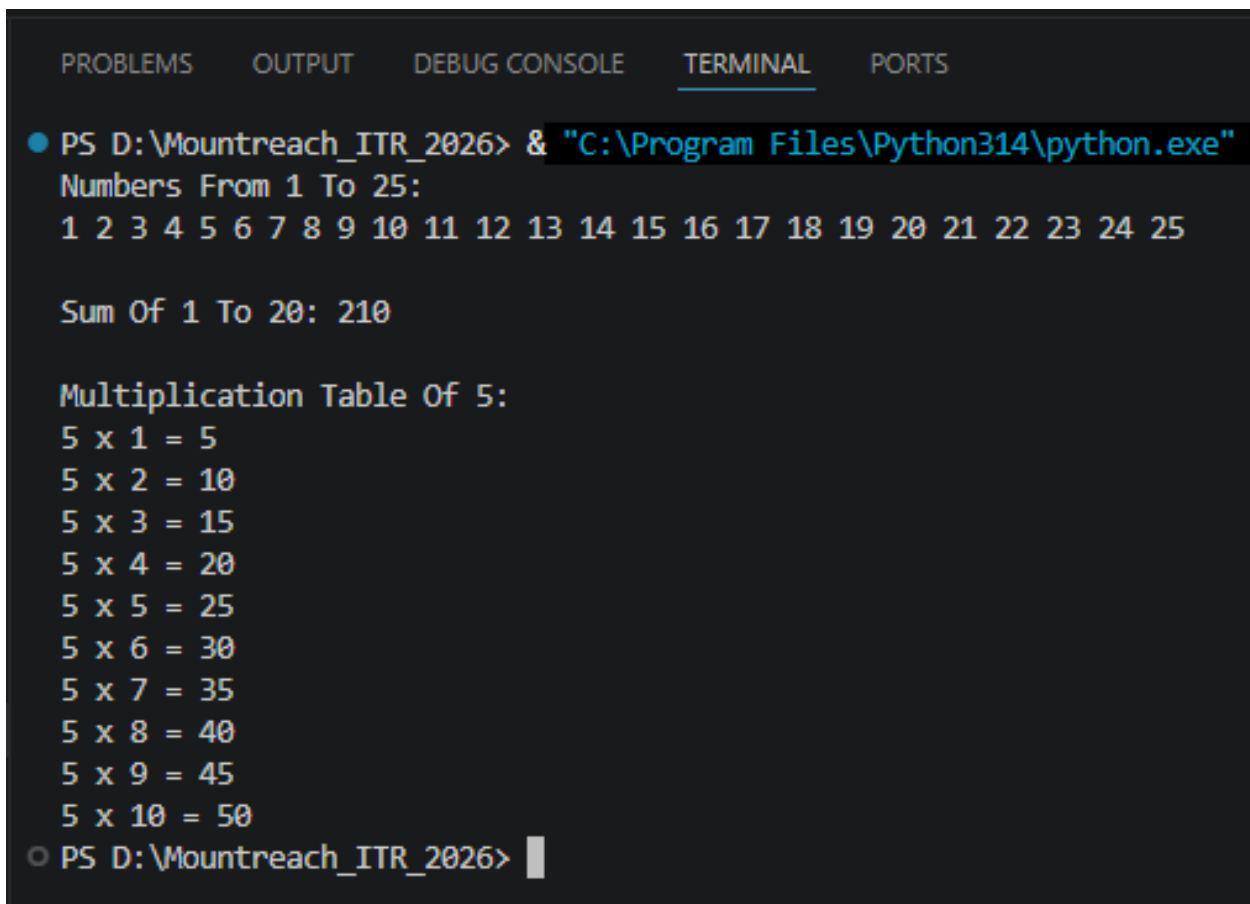
3. For Loop Revision

Write a program using a for loop and range() to:

- Print all numbers from 1 to 25.
- Print the sum of all numbers from 1 to 20.
- Print the table of 5 5 x 1 to 5 x 10.

Add comments explaining when to use for loop.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Numbers From 1 To 25:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Sum Of 1 To 20: 210

Multiplication Table Of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
print("Numbers From 1 To 25:")
for i in range(1, 26):          # Starts At 1, Stops Before 26
    print(i, end=" ")

total = 0
for i in range(1, 21):         # Starts At 1, Stops Before 21
```

```
    total += i
print("\n\nSum Of 1 To 20:", total)

print("\nMultiplication Table Of 5:")
for i in range(1, 11):      # Starts At 1, Stops Before 11
    print("5 x", i, "=", 5 * i)

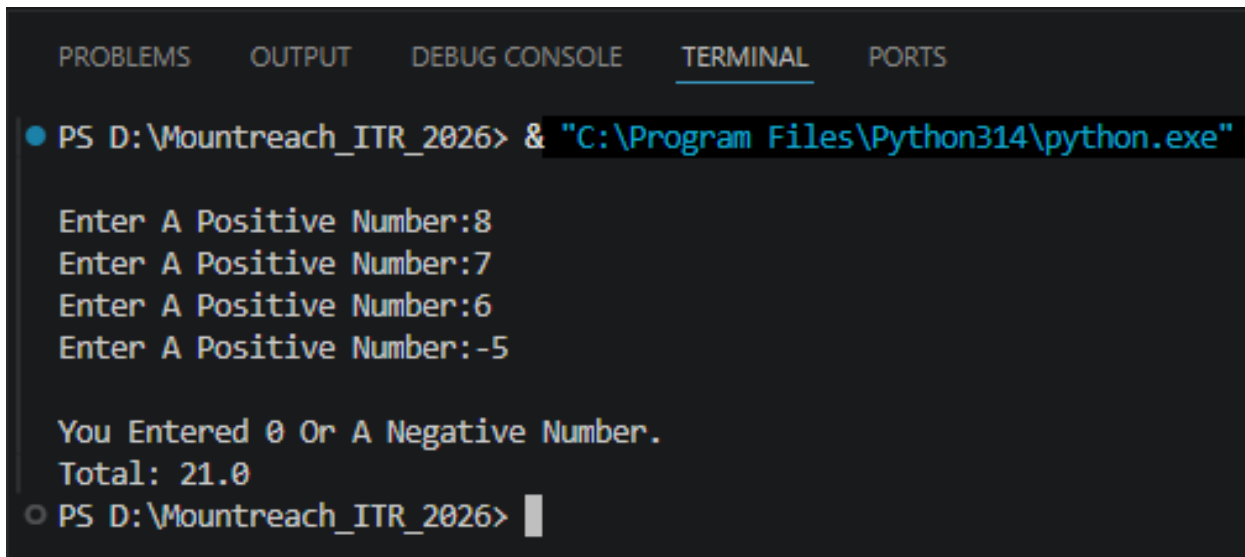
# When To Use A For Loop:
# - You Know In Advance How Many Iterations Are Needed.
# - You Want To Iterate Over A Sequence (List, Range, String, etc.).
# - Cleaner & Safer Than While When The Count Is Fixed.
```

4. While Loop Revision

Write a program using a while loop that:

- Repeatedly asks the user to enter a positive number.
- Keeps adding the numbers to a running total.
- Stops when the user enters 0 or a negative number.
- Finally prints the total sum.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
• PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

Enter A Positive Number:8
Enter A Positive Number:7
Enter A Positive Number:6
Enter A Positive Number:-5

You Entered 0 Or A Negative Number.
Total: 21.0
• PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
# Initialize The Total To 0
total = 0

# Take Input From The User For The First Number Before Entering The Loop
number = float(input("Enter A Positive Number:"))

# Keep Looping Until The User Enters A Positive Number
while number > 0:

    # Add The Entered Number To The Total
    total = total + number

    # Ask The User For The Next Number
    number = float(input("Enter A Positive Number:"))

print("\nYou Entered 0 Or A Negative Number.")
print("Total:", total)
```

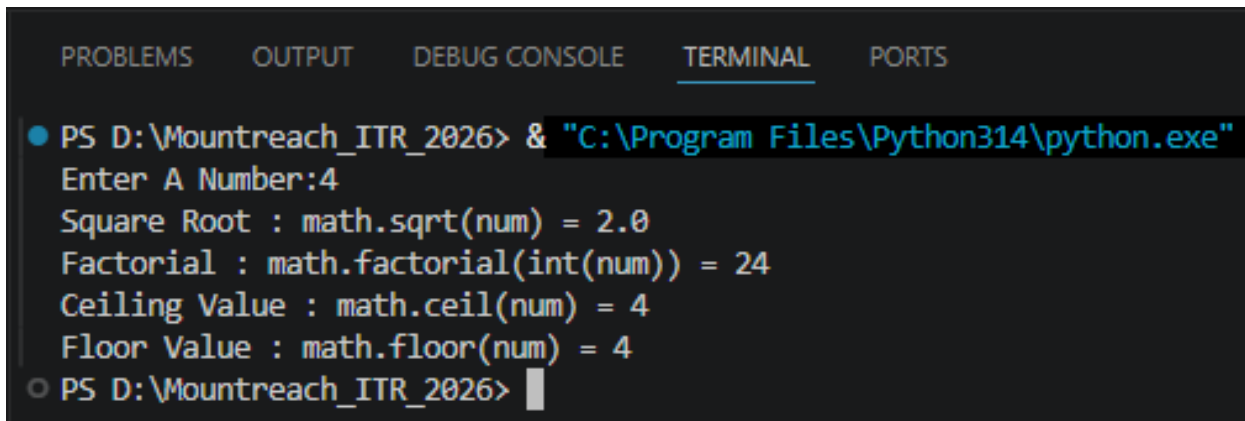
5. Math Module

Write a program that imports the math module and does the following:

- Takes a number as input.
- Prints its square root using `math.sqrt()`.
- Prints its factorial using `math.factorial()` (for positive integers only).
- Prints the ceiling and floor value using `math.ceil()` and `math.floor()`.

Add comments about the usefulness of the math module.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A Number:4
Square Root : math.sqrt(num) = 2.0
Factorial : math.factorial(int(num)) = 24
Ceiling Value : math.ceil(num) = 4
Floor Value : math.floor(num) = 4
○ PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
import math # Built-In Module With Mathematical Functions & Constants

num = float(input("Enter A Number:"))

print("Square Root : math.sqrt(num) =", math.sqrt(num))

if num >= 0 and num == int(num): # Factorial Only Works On Non-Negative Integers
    print("Factorial : math.factorial(int(num)) =", math.factorial(int(num)))
else:
    print("Factorial : Only Defined For Non-Negative Integers")

print("Ceiling Value : math.ceil(num) =", math.ceil(num))
print("Floor Value : math.floor(num) =", math.floor(num))
```

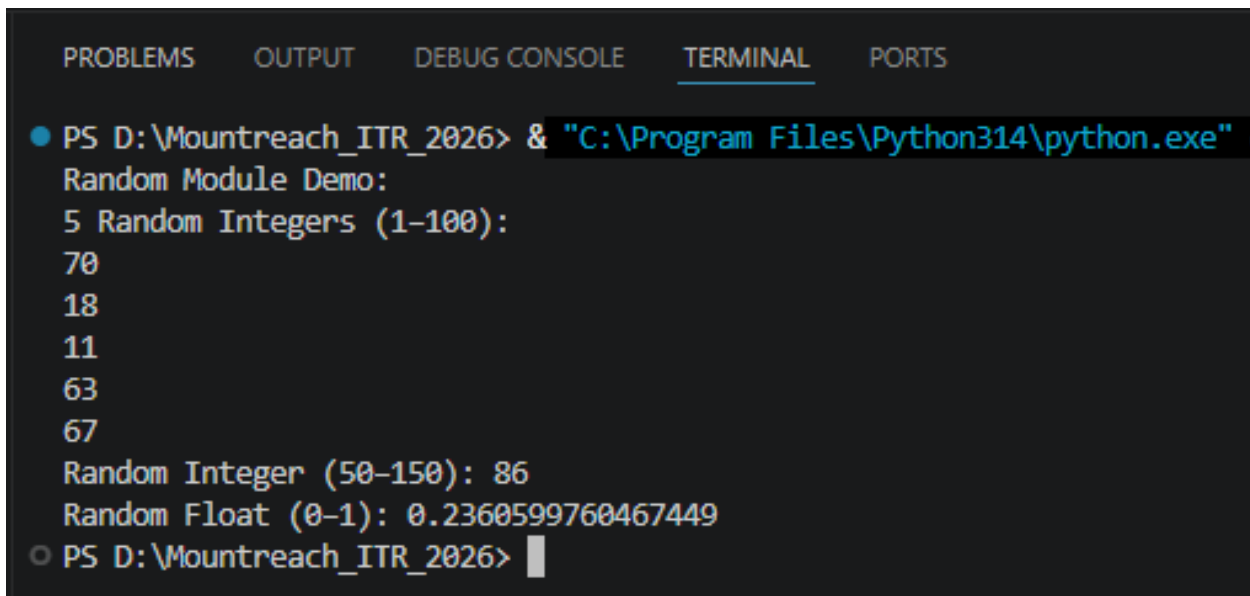
6. Random Module

Write a program using the random module that:

- Generates and prints 5 random integers between 1 and 100.
- Generates and prints one random number between 50 and 150.
- Prints a random floating point number between 0 and 1.

Use different functions from the random module.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Random Module Demo:
5 Random Integers (1-100):
70
18
11
63
67
Random Integer (50-150): 86
Random Float (0-1): 0.2360599760467449
○ PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
import random # Built-In Module For Generating Pseudo-Random Numbers

print("Random Module Demo:")

# Generate 5 Random Integers Between 1 & 100
print("5 Random Integers (1-100):")
for _ in range(5):
    print(random.randint(1, 100))

# One Random Integer Between 50 & 150
print("Random Integer (50-150):", random.randrange(50, 150))

# Random Float Between 0 & 1
print("Random Float (0-1):", random.random())
```

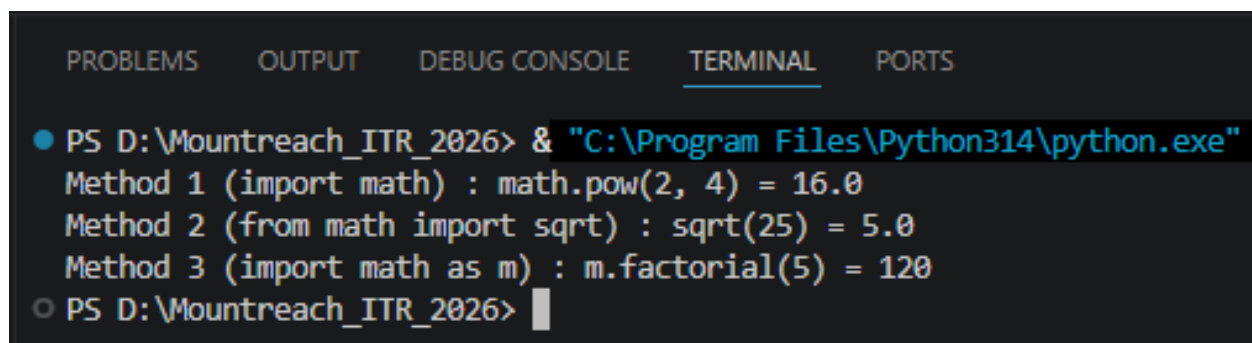
7. Import Methods

Demonstrate different ways of importing modules:

- Import the entire math module using `import math` and use `math.pow(2, 4)`.
- Import only `sqrt` from `math` using `from math import sqrt`.
- Import `math` with an alias as `m` and use `m.factorial(5)`.

Print the results with clear messages.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Method 1 (import math) : math.pow(2, 4) = 16.0
Method 2 (from math import sqrt) : sqrt(25) = 5.0
Method 3 (import math as m) : m.factorial(5) = 120
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
# Importing Module
import math
print("Method 1 (import math) : math.pow(2, 4) =", math.pow(2, 4))

# Import Specific Function - Use Directly Without Prefix
from math import sqrt
print("Method 2 (from math import sqrt) : sqrt(25) =", sqrt(25))

# Import With Alias - Shorter Prefix, Useful For Long Module Names
import math as m
print("Method 3 (import math as m) : m.factorial(5) =", m.factorial(5))
```

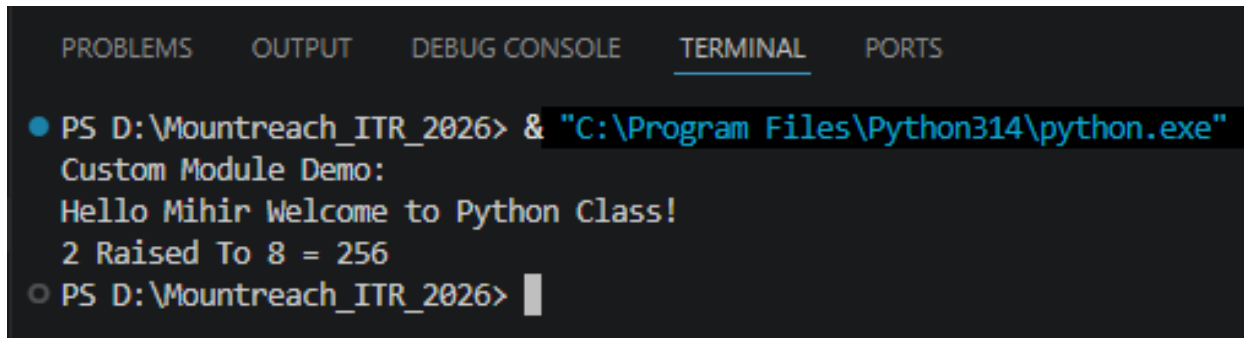

8. Custom Module

Create a custom module file named `mymodule.py` that contains:

- A function `greet(name)` that prints "Hello [name], Welcome to Python Class!"
- A function `calculate_power(base, exp)` that returns base raised to exponent.

Then write a main program that imports this module and uses both functions.

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal shows a command prompt where the user has run a Python command. The output of the command is displayed in a light blue font. The output consists of three lines: 'Custom Module Demo:', 'Hello Mihir Welcome to Python Class!', and '2 Raised To 8 = 256'. The prompt then returns to the command line.

```
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Custom Module Demo:
Hello Mihir Welcome to Python Class!
2 Raised To 8 = 256
PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
#mymodule.py
def greet(name):
    print("Hello", name, "Welcome to Python Class!")

def calculate_power(base, exp):
    return base ** exp

#main.py
import mymodule as mm

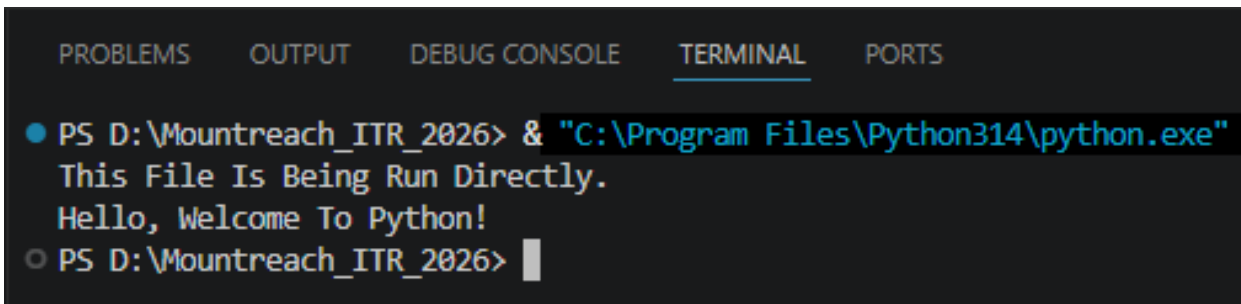
print("Custom Module Demo:")
mm.greet("Mihir")
print("2 Raised To 8 =", mm.calculate_power(2, 8))
```

9. name == "main"

Write a program that:

- Creates a small function and demonstrates the use of `if __name__ == "__main__":` so that some code runs only when the file is executed directly.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  This File Is Being Run Directly.
  Hello, Welcome To Python!
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
# Define Function
def greet():
    print("Hello, Welcome To Python!")

# This Condition Checks Whether The Script Runs Directly
# If True, The Code Inside This Block Will Execute
# If The File Is Imported Into Another File, This Block Will Not Run
if __name__ == "__main__":
    print("This File Is Being Run Directly.")

    # Call The Greet Function
    greet()
```

10. Mini Project - Combined Concepts

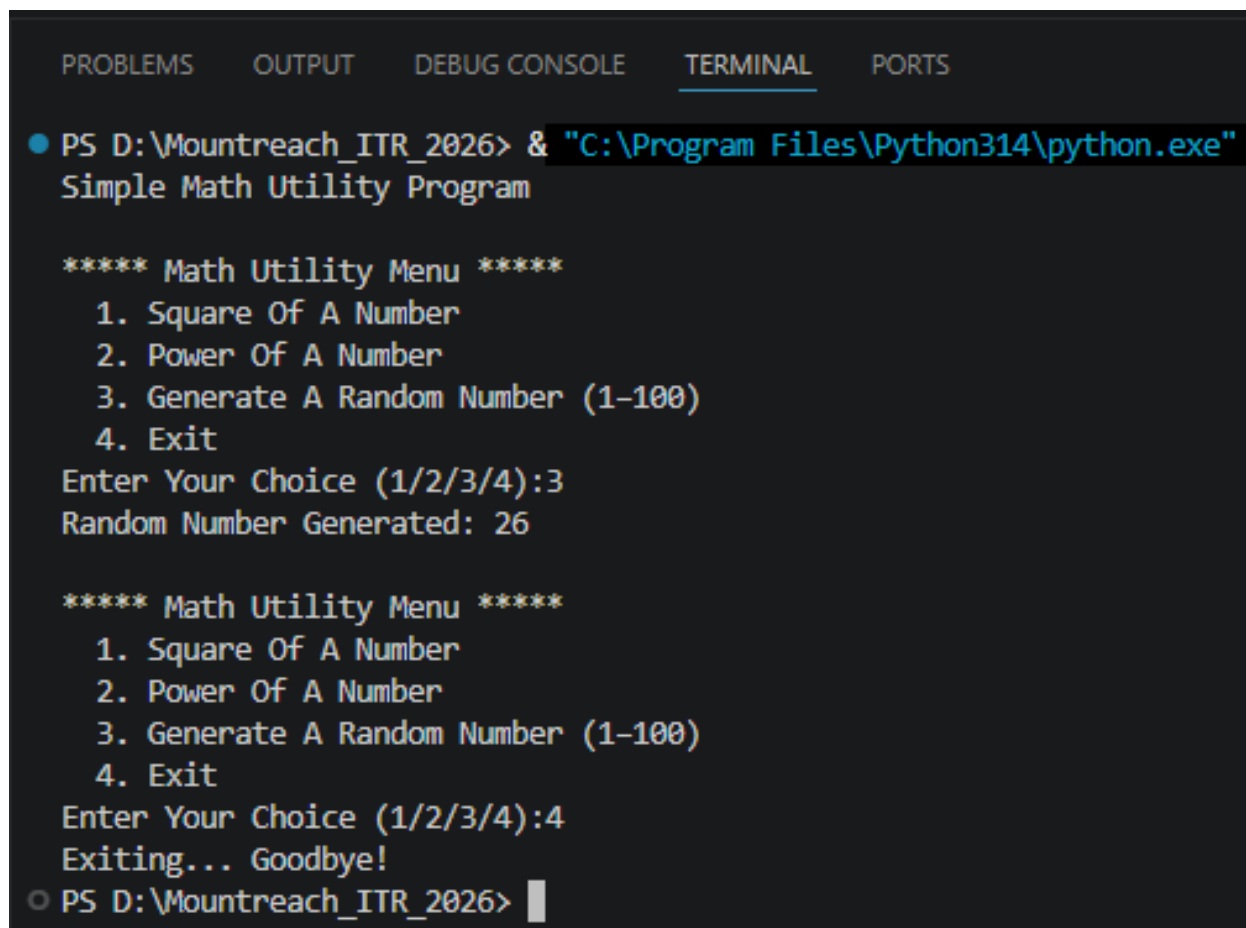
Create a Simple Math Utility Program using functions, lambda, loops and modules:

Write the following:

- A lambda function to calculate square of a number.
- A normal function using math module to calculate power.
- Use a while loop to repeatedly ask the user for a number and choice Square / Power / Random Number).
- Use random module to generate a random number when chosen.

Continue until the user chooses to exit. Use proper functions and comments.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Simple Math Utility Program

***** Math Utility Menu *****
1. Square Of A Number
2. Power Of A Number
3. Generate A Random Number (1-100)
4. Exit
Enter Your Choice (1/2/3/4):3
Random Number Generated: 26

***** Math Utility Menu *****
1. Square Of A Number
2. Power Of A Number
3. Generate A Random Number (1-100)
4. Exit
Enter Your Choice (1/2/3/4):4
Exiting... Goodbye!
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
import math
import random
```

```

# Lambda Function
square = lambda x: x ** 2

# Normal Functions
def calculate_power(base, exp):
    return math.pow(base, exp)

def display_menu():
    print("\n***** Math Utility Menu *****")
    print("  1. Square Of A Number")
    print("  2. Power Of A Number")
    print("  3. Generate A Random Number (1-100)")
    print("  4. Exit")

print("Simple Math Utility Program")

while True:
    display_menu()
    choice = input("Enter Your Choice (1/2/3/4):")

    if choice == "1":
        num = float(input("Enter A Number:"))
        print("Square Of", num, "=", square(num))

    elif choice == "2":
        base = float(input("Enter Base:"))
        exp = float(input("Enter Exponent:"))
        print(base, "^", exp, "=", calculate_power(base, exp))

    elif choice == "3":
        rand_num = random.randint(1, 100)
        print("Random Number Generated:", rand_num)

    elif choice == "4":
        print("Exiting... Goodbye!")
        break

    else:
        print("Invalid Choice. Please Enter 1, 2, 3, Or 4.")

```